

Milestone 4 Technical Summary

System goal. Help a graduate business career advisor screen job opportunities for MSBA international students. The prototype takes an advisor query or raw posting text, retrieves similar real postings, predicts a triage label, and generates a grounded recommendation.

Architecture

Input. Advisor query or job posting text. **Retriever.** Sparse inverted index over 80,000 training postings from the 100,000-row real project sample. **Decision layer.** Transparent MSBA role-fit rubric; retrieved examples ground the answer. **Generation.** Template output that cites retrieved title, company, location, skills, label, and authorization caveat. **Output.** Recommendation plus evidence list for advisor review.

Stage	Prototype behavior
Input	advisor/student query or raw job posting text
Preprocess	normalize title, skills, location, schedule, and posting text
Retrieve	sparse inverted index over 80,000 training postings
Decide	apply a transparent role-fit rubric; retrieval supplies grounding and comparable examples
Generate	template output that cites retrieved posting fields and flags missing authorization evidence
Evaluate	accuracy, macro F1, retrieval success, and support hit@5 against weak labels

Evaluation vs. Baseline

System	Eval set	Accuracy	Macro F1
Milestone 2 published TF-IDF baseline	20k validation	79.3%	78.5%
Same-subset hashed centroid baseline	4k validation	79.7%	79.1%
Milestone 4 retrieval + grounded prototype	4k validation	80.0%	80.1%

The prototype improves same-subset macro F1 from 79.1% to 80.1%. Retrieval also provides evidence for generated output: retrieval success rate is 100.0%, and support hit@5 is 90.0%. The comparison is preliminary because the labels are weak labels, not human advisor labels.

Grounded Output Example

Input query: Find entry-level US analytics or business analyst opportunities for MSBA international students with SQL, Python, Excel, Tableau, or Power BI skills. Flag cases where authorization evidence is missing.

System headline: Review 3 of the retrieved postings first; they match analytics role-fit signals.

Entry level / Business Data Analyst (Remote) (high_fit). title/location: Entry level / Business Data Analyst (Remote) at Lumos Stratgy in San Francisco, CA; skills: sql, python, r, excel, tableau, power bi; authorization evidence is not available in this public source

Entry Level Data Analyst (high_fit). title/location: Entry Level Data Analyst at SkiHomeRealty in Anywhere; skills: sql, python, r, tableau, power bi; authorization evidence is not available in this public source

Entry Level Business Analyst/Data Analyst (high_fit). title/location: Entry Level Business Analyst/Data Analyst at Alipro in Tysons, VA; skills: sql, python, aws, excel, tableau, flow; authorization evidence is not available in this public source

Entry Level Data Analyst - US Army (13J) (unclear). title/location: Entry Level Data Analyst - US Army (13J) at United States Army in Long Beach, CA; skills: skills not listed; authorization evidence is not available in this public source

Risk and Grounding Analysis

Risk 1: hallucinated authorization claims. The dataset does not contain reliable CPT, OPT, or sponsorship text. Mitigation: template generation never claims eligibility; it explicitly says authorization evidence is missing and requires advisor review.

Risk 2: overconfidence from weak labels. High_fit, medium_fit, low_fit, and unclear are rule-derived labels. Mitigation: the output shows retrieved evidence and label reason, and the system is framed as advisor support, not an automatic forwarding or rejection tool.

Next step: add de-identified Career Center postings that include authorization language, then evaluate answer faithfulness and high-fit precision with human-reviewed labels.